

Principal Component Analysis

Statistical Data Mining II

Spring 2022

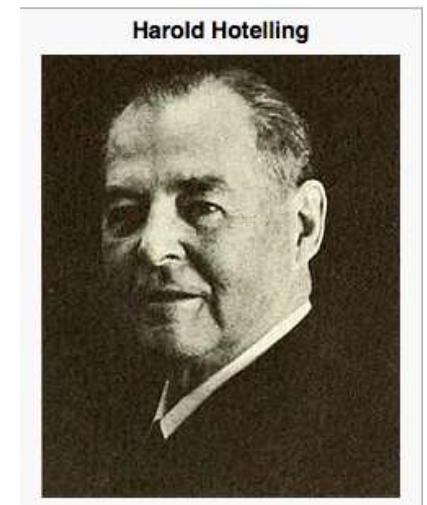
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Outline

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- Least-Squares Optimality of PCA
- PCA as a Variance Maximization Technique
- Sample Principal Components
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- Example: Face Recognition
- Example: Handwriting
- What can we gain from PCA?

A brief history

- Origins somewhat difficult to trace
- Singular Value Decomposition (SVD)
Independently discovered by Beltrami (1873) and Jordan (1874).
- Applied SVD
Fisher and Mackenzie (1923) used the SVD in the context of an agricultural trial.
- Earliest descriptions of PCA
Pearson (1901) and Hotelling (1933)



A brief history

Hotelling's Motivation:

There may be a smaller “fundamental set of independent variables... which determine the values” of the original p variables.

- Sometimes called “factors” or alternatively “components”.
- Hotelling “chooses” his components to maximize their successive to the total variances of the original variables, and calls the components derived this way, principal components.
- The analysis which identifies these components is termed “principal component analysis” (PCA).

A brief history

In his original paper:

- Principal components are found with the power method.
- Discusses a different interpretation in terms of ellipsoids and constant probability for multivariate normal distributions.
- *Little was done with the PCA for the 25 years that followed publication. One limitation was computational power. It was not feasible to do PCA at the time for a p greater than 4. But this was exactly when you would want to do PCA.*
- *Methods for computing the SVD emerged in the 1950's, and were refined in the 1960's.*
- *Explosion since then!*

Introduction

- **Objective:** derive a reduced set of orthogonal linear projections of a single collection of correlated variables,

$$X = (X_1, X_2, \dots, X_r)^T$$

where the projections are ordered by decreasing variances.

- PCA also referred to as a method for decorrelating -- used in many other fields disguised under different names, e.g, *Karhunen-Loeve transform (communications theory)* and *empirical orthogonal functions (atmospheric sciences)*.
- Can be used in the supervised and unsupervised setting.

Introduction

- Aside from “dimension reduction” and “de-correlating”, PCA can be used for discovery.
- Discovery takes the form of graphical displays of the principal component scores. The **first few principal component** scores can reveal whether most of the data actually live on a linear subspace, and can be used to:
 - flag outliers
 - discovery anomalies in the distribution
 - identify groupings/ clusters
 - distinguish coordination with experimental treatment factors
- The **last few principal components** also useful from the point of view of outlier detection.

Motivating Example

The Nutritional Value of Food

- 961 food items.
- Nutritional components of each food item are given by the following seven variables: fat (grams), food energy (calories), carbohydrates (grams), protein (grams), cholesterol (milligrams), weight (grams) and saturated fat (grams).
- Food items are listed according to serving sizes, which vary, cup, loaf, piece etc.
- To standardize the different types of servings, the data is scaled and normalized.

```
food {MMST}
```

```
MMST FOOD DATA
```

```
Description
```

```
nutritional value of food, 196, 198, 206, 208, 462, 612, 613, 631
```

```
Usage
```

```
data(food)
```

```
Format
```

```
Print
```

```
A data frame with 961 observations on the following 7 variables.
```

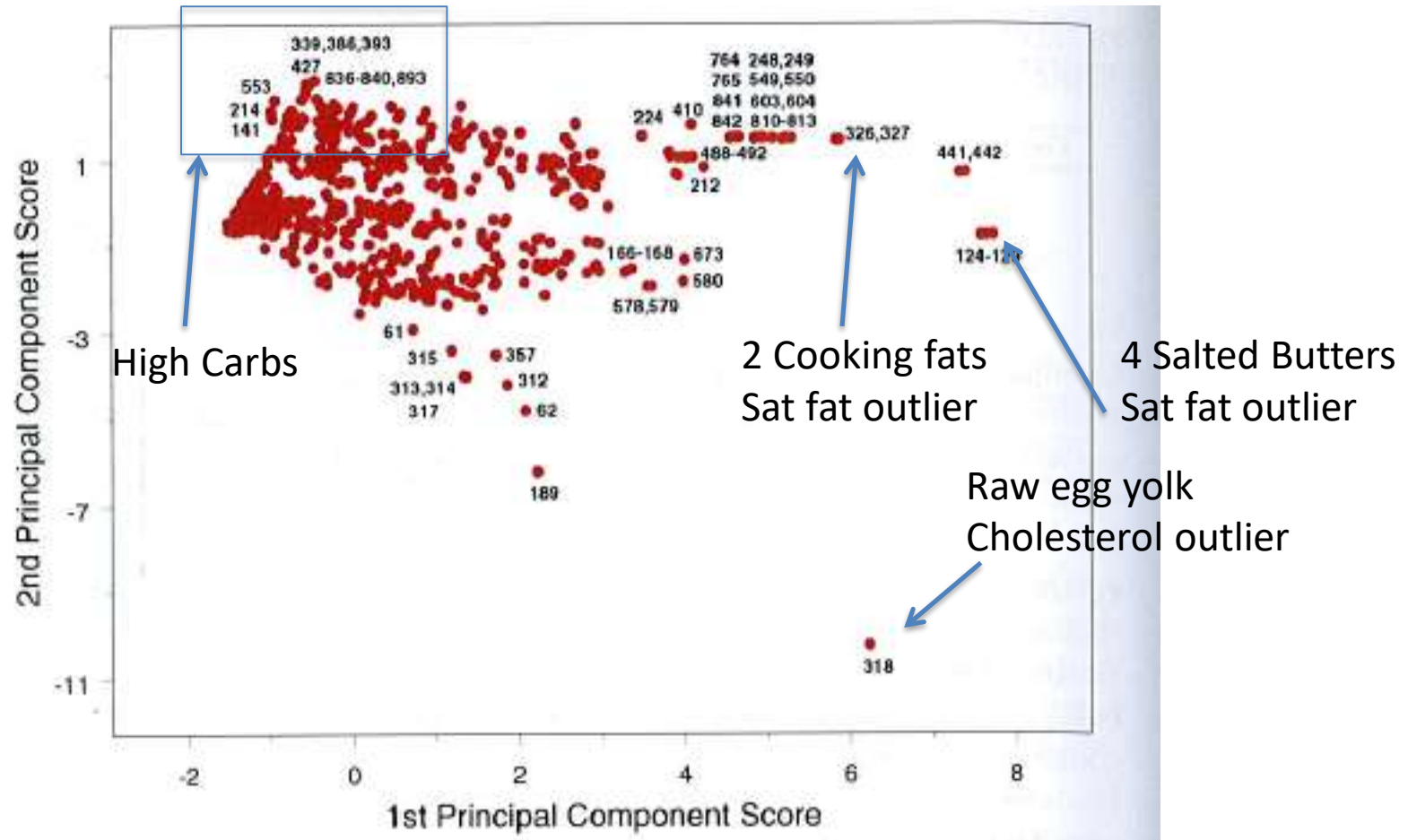

Motivating Example

Before: large matrix 961 x 6

After:

Food Component	PC1	PC2	PC3	PC4	PC5	PC6
Fat	0.557	0.099	0.275	0.130	0.455	0.617
Food energy	0.536	0.357	-0.137	0.075	0.273	-0.697
Carbohydrates	-0.025	0.672	-0.568	-0.286	-0.157	0.344
Protein	0.235	-0.374	-0.639	0.599	-0.154	0.119
Cholesterol	0.253	-0.521	-0.326	-0.717	0.210	-0.003
Saturated fat	0.531	-0.019	0.261	-0.150	-0.791	0.022
Variance	2.649	1.330	1.020	0.680	0.267	0.055
% Total Variance	44.1	22.2	17.0	11.3	4.4	0.9

Motivating Example



Principal Components

General formulation (ISLR):

Let $Z_1, Z_2, \dots, Z_M$ represent $M < p$ linear combinations of our original p predictors. That is,

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j \quad (6.16)$$

for some constants $\phi_{1m}, \phi_{2m}, \dots, \phi_{pm}$, $m = 1, \dots, M$. We can then fit the linear regression model

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i, \quad i = 1, \dots, n, \quad (6.17)$$

using least squares. Note that in (6.17), the regression coefficients are given by $\theta_0, \theta_1, \dots, \theta_M$. If the constants $\phi_{1m}, \phi_{2m}, \dots, \phi_{pm}$ are chosen wisely, then such dimension reduction approaches can often outperform least squares

Principal Components

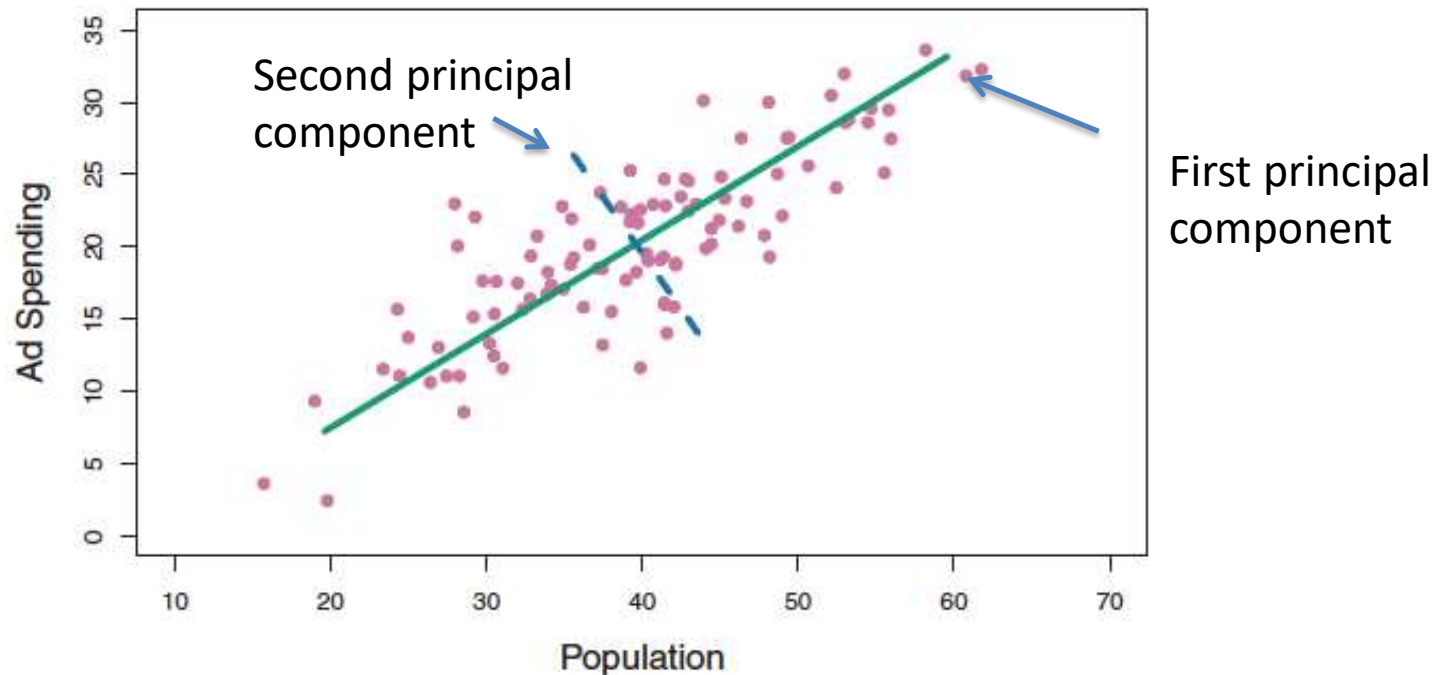


FIGURE 6.14. The population size (**pop**) and ad spending (**ad**) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

Principal Components

What is meant by “projecting data”?

- The green line represents the the first principal component direction of the data... greatest variability in the data.
- If we projected 100 observations onto this line, then these would have the most variance (over any competing line).
- The blue line is the second principal component direction, which contains the most information in the data, subject to the fact it has to be orthogonal to PC1.

Projecting a point onto a line →
Finding the location on the line which is
Closest to the point!

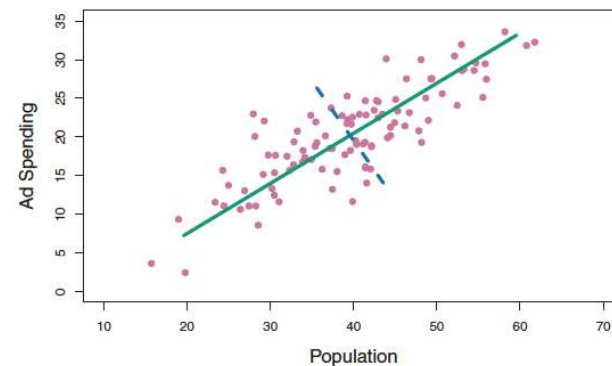


FIGURE 6.14. The population size (*pop*) and ad spending (*ad*) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

Principal Components

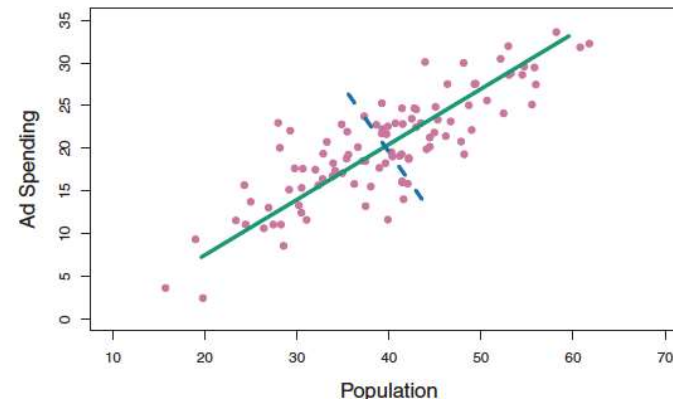
Mathematically PC1:

$$Z_1 = 0.839 \frac{\text{pop} - \overline{\text{pop}}}{\sigma_{\text{pop}}} + 0.544 \frac{\text{ad} - \overline{\text{ad}}}{\sigma_{\text{ad}}}$$

Loadings f_i

Note that

$$\sum_{i=1}^M f_i^2 = 1$$



*This particular linear combination
Yields the maximum possible variance,
of all linear combinations!

FIGURE 6.14. The population size (**pop**) and ad spending (**ad**) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

Principal Components

Mathematically data points projected to PC1 and PC2:

$$z_{i1} = 0.839 \left(\frac{ad_i - \overline{ad}}{\sigma_{ad}} \right) + 0.544 \left(\frac{pop_i - \overline{pop}}{\sigma_{pop}} \right)$$

$$z_{i2} = 0.544 \left(\frac{ad_i - \overline{ad}}{\sigma_{ad}} \right) - 0.839 \left(\frac{pop_i - \overline{pop}}{\sigma_{pop}} \right)$$

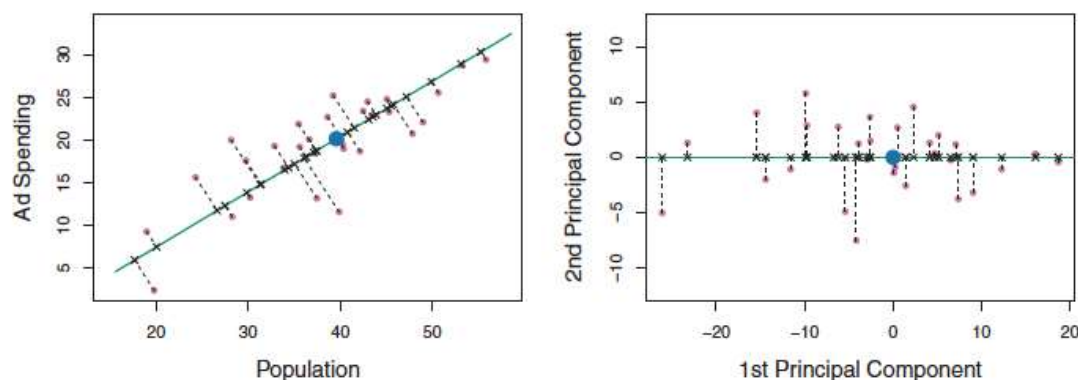


FIGURE 6.15. A subset of the advertising data. The mean **pop** and **ad** budgets are indicated with a blue circle. Left: The first principal component direction is shown in green. It is the dimension along which the data vary the most, and it also defines the line that is closest to all n of the observations. The distances from each observation to the principal component are represented using the black dashed line segments. The blue dot represents $(\overline{pop}, \overline{ad})$. Right: The left-hand panel has been rotated so that the first principal component direction coincides with the x-axis.

Principal Components

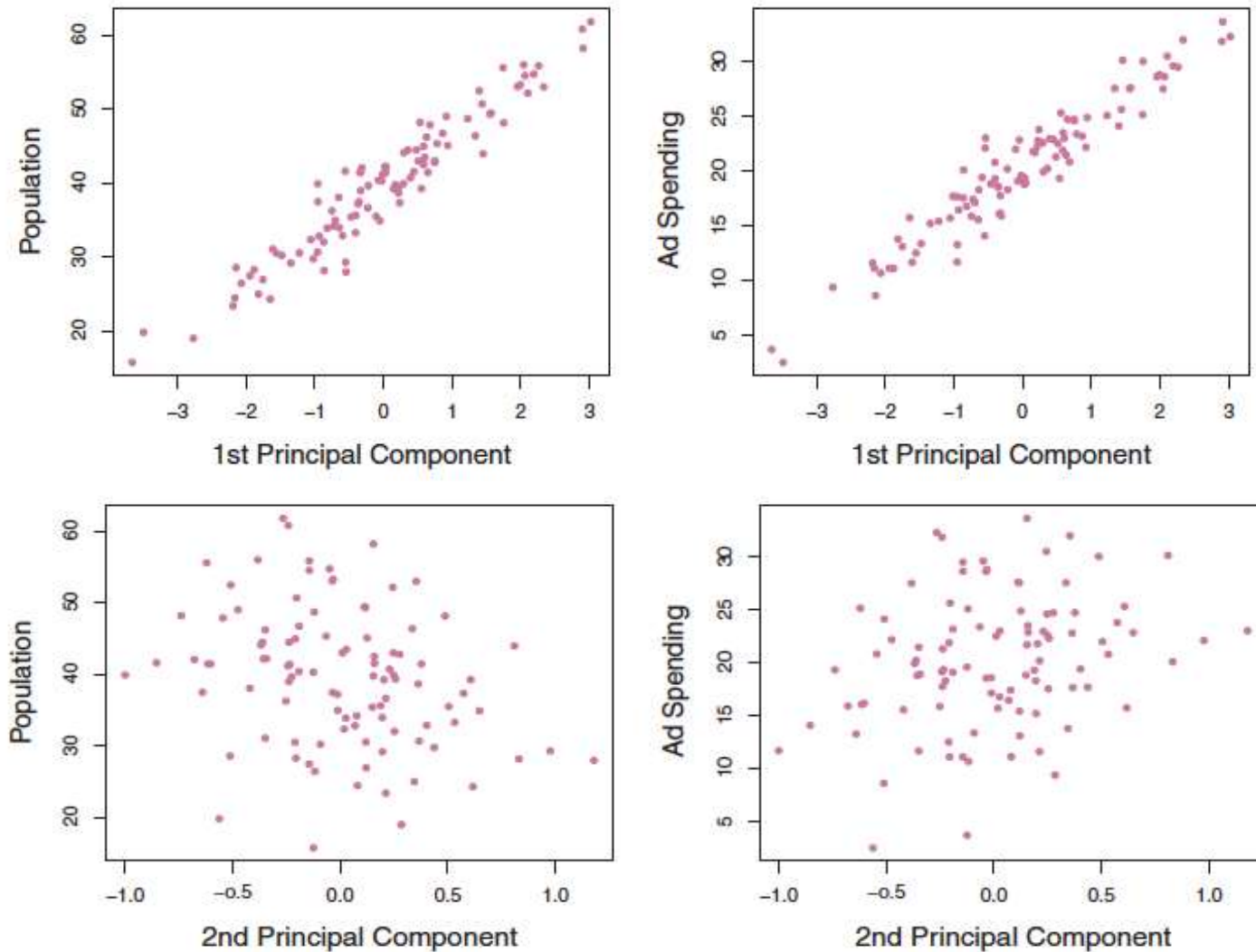


FIGURE 6.17. *Plots of the second principal component scores z_{i2} versus **pop** and **ad**. The relationships are weak.*

Population Principal Components

- Assume that the random r - vector:

$$X = (X_1, \dots, X_r)^T$$

has mean $\boldsymbol{m}_X$ and $(r \times r)$ covariance matrix $\boldsymbol{S}_{XX}$.

- PCA seeks to replace the set of r (unordered and correlated) input variables $X_1, \dots, X_r$ by a smaller set of t (ordered and uncorrelated) linear projections, $x_1, \dots, x_t$ for $t \leq r$ of the input variables:

$$x_j = \boldsymbol{b}_j' X = b_{j1}X_1 + \dots + b_{jr}X_r, \quad j = 1, 2, \dots, t,$$

where we minimize the loss of information due to replacement.

The first t principal components of X

Population Principal Components

- In PCA “information” is interpreted as the “total variation” of the original input variables:

$$\sum_{j=1}^r \text{var}(X_j) = \text{trace}(\mathbf{S}_{XX}).$$

- We can decompose $\mathbf{S}_{XX}$ using an eigenvalue decomposition:

$$\mathbf{S}_{XX} = \mathbf{U}\mathbf{D}\mathbf{U}^T, \quad \mathbf{U}^T\mathbf{U} = \mathbf{I}_r,$$

where the diagonal matrix $\mathbf{D}$ has diagonal elements and the eigenvalues, $\{\lambda_j\}$, of $\mathbf{S}_{XX}$, and the columns of $\mathbf{U}$ are the eigenvectors $\mathbf{S}_{XX}$.

- The total variation is $\text{tr}(\mathbf{S}_{XX}) = \text{tr}(\mathbf{D}) = \sum_{j=1}^r \lambda_j$.

Population Principal Components

The j th coefficient vector $b_j = (b_{1j}, b_{2j}, \dots, b_{2j})^T$, is chosen so that:

- The first t linear projections X_j , $j=1, 2, \dots, t$, of X are ranked in importance through their variances $\{\text{var}\{X_j\}\}$, which are listed in decreasing order of magnitude: $\text{var}\{X_1\} \geq \text{var}\{X_2\} \geq \dots \geq \text{var}\{X_t\}$
- X_j is uncorrelated with all X_k , $k < j$.

Two popular derivations:

1. Least squares optimality criterion
2. Variance maximization

Least Squares Optimality of PCA

We want to find an r -vector m and an $(r \times t)$ matrix A such that the projections X have the property that $X = m + AX$ some least squares sense.

- Least squares criterion (minimize):

$$E\left\{(X - m - AX)^T (X - m - AX)\right\},$$

is a measure of how well we reconstruct by the projection.

$$X = BX$$

- We can reformulate the criterion as:

$$E\left\{(X - \underline{m} - \underline{ABX})^T (X - \underline{m} - \underline{ABX})\right\}.$$

Find unknowns which Minimize.

Least Squares Optimality of PCA

The least squares criterion:

$$E\left\{\left(X - m - ABX\right)^T \left(X - m - ABX\right)\right\},$$

Is minimized with a reduced-rank regression.

Specifically:

$$A^{(t)} = (v_1, \dots, v_t) = B^{(t)T}$$

$$m^{(t)} = (I_r - A^{(t)}B^{(t)})m_X$$

where $v_j = v_j(S_{XX})$ is the eigenvector associated with the j th largest eigenvalue, λ_j , of S_{XX} .

Least Squares Optimality of PCA

Therefore, our best “rank t ” approximation to the original X is given by:

$$\begin{aligned}\hat{X}^{(t)} &= \bar{m}^{(t)} + C^{(t)} X \\ &= \bar{m}_X + C^{(t)} (X - \bar{m})\end{aligned}$$

where,

$$C^{(t)} = A^{(t)} B^{(t)} = \sum_{j=1}^t v_j v_j^T$$

can be viewed as the reduced rank regression coefficient matrix with rank t .

The minimum of: $E\left\{(X - \bar{m} + ABX)^T (X - \bar{m} + ABX)\right\}$, is given by the sum of the smallest $r-t$ eigenvalues of S_{XX} .

Least Squares Optimality of PCA

The least squares criterion:

$$\star E\left\{\left(X - m + ABX\right)^T \left(X - m + ABX\right)\right\},$$

Is minimized with a reduced-rank regression.

Specifically:

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where $v_j = v_j(S_{XX})$ is the eigenvector associated with the j th largest eigenvalue, λ_j , of S_{XX} .

Least Squares Optimality of PCA

Principal Components:

- The first t principal components of X are given by the linear projections: $x_j = v_j^T X$, $j = 1, \dots, t$.

- Note that these are “orthogonal projections” therefore:

$$\text{cov}(x_i, x_j) = \text{cov}(v_i^T X, v_j^T X) = v_i^T S_{XX} v_j = \lambda_j v_i^T v_j = d_{ij} \lambda_j.$$

where d_{ij} is 1 if $i = j$ and zero otherwise.

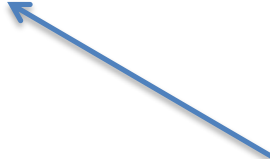
- λ_1 , the largest eigenvalue of S_{XX} is $\text{var}(x_1)$.
- λ_2 , the second largest eigenvalue of S_{XX} is $\text{var}(x_2)$.
- ... and so on.

Least Squares Optimality of PCA

Goodness of Fit:

Proportion of the total variation in the input variables that is explained by the last $r-t$ principal components:

$$\frac{\lambda_{r+1} + \dots + \lambda_t}{\lambda_1 + \dots + \lambda_r}$$



Should be small, if the first r components do a good job capturing the variation of X .

Sample Principal Components

Back to reality:

- We have a population, which is represented by our data, and is finite.
- We have to estimate principal components using n independent observations, on $\{X_i, i=1, \dots, n\}$, on X .
- We have to estimate the mean and covariance as well, via sample mean and sample covariance.
- In this case, we are interested in $\hat{S}_{XX}$.

Sample Principal Components

- The ordered eigenvalues of $\hat{\mathbf{S}}_{XX}$ are denoted by:

$$\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_r \geq 0$$

and the eigenvector $\hat{\mathbf{v}}_j$ corresponds to the j th eigenvalue.

- Similar to what we did before, we estimate:

$$\hat{\mathbf{A}}^{(t)} = (\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_t) = \hat{\mathbf{B}}^{(t)T}$$

where $\hat{\mathbf{v}}_j$ is the j th sample eigenvector of $\hat{\mathbf{S}}_{XX}$, $j = 1, 2, \dots, t$, and $t \leq r$.

Sample Principal Components

What is the best rank- t reconstruction of X ?

- The “best” low rank approximation is of the form:

$$\hat{X} = \bar{X} + \hat{C}^{(t)}(X - \bar{X})$$

where

$$\hat{C}^{(t)} = \hat{A}^{(t)} \hat{B}^{(t)} = \mathbf{\hat{a}} \sum_{j=1}^t \hat{v}_j \hat{v}_j^T$$

can be viewed as the reduced-rank regression coefficient matrix.

- The j th sample PC score of X is given as:

$$\hat{x}_j = \hat{v}_j^T X_C$$

where $X_C = X - \bar{X}$.

Sample Principal Components

Goodness of Fit:

Proportion of the total variation in the input variables that is explained by the last $r-t$ principal components:

$$\frac{\hat{I}_{r+1} + \dots + \hat{I}_t}{\hat{I}_1 + \dots + \hat{I}_r}$$



Should be small, if the first t components do a good job capturing the variation of X .

**Generally one hopes that the sample variances of the first few principal components will be large, and the rest will be small enough for the corresponding set of sample PCs to be omitted.*

**Rationale: a variable (PC) that does not change much relative to others, may be treated as a constant and omitted.*

Sample Principal Components

- Some asymptotic results exist for (1939 Fisher et al. 1951 Mood)
- When the dimensionality of r is large (even larger than n), there are no such results.
- Random matrix theory, has produced some convergence results e.g., Quarter circle law (Marcenko and Pastur, 1967).

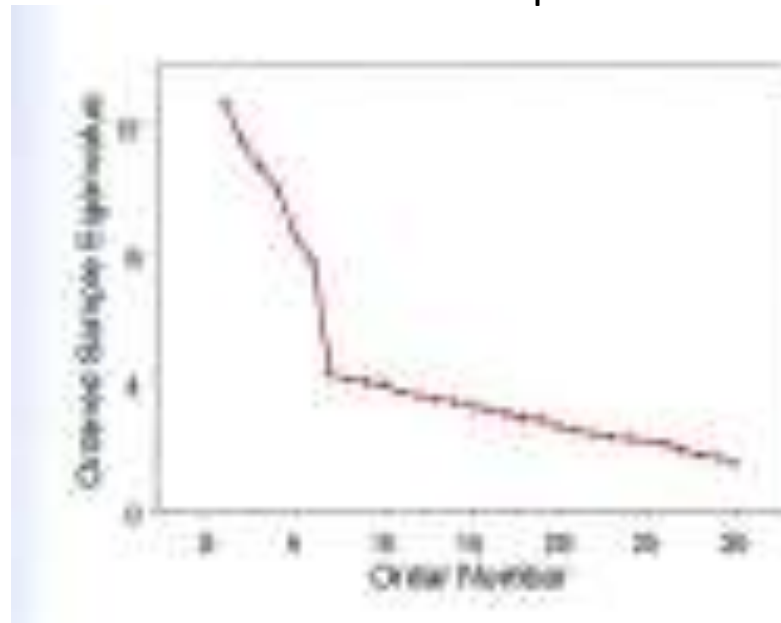
How many principal components to retain?

- **Recall:** the criterion for a good projection in PCA is a high variance for that projection, we should retain only those principal components with large variances.
- Boils down to looking at the magnitude of the eigenvalues, of $\hat{S}_{XX}$ and asking:
How small can an eigenvalue be while still regarding the corresponding principal components as significant?

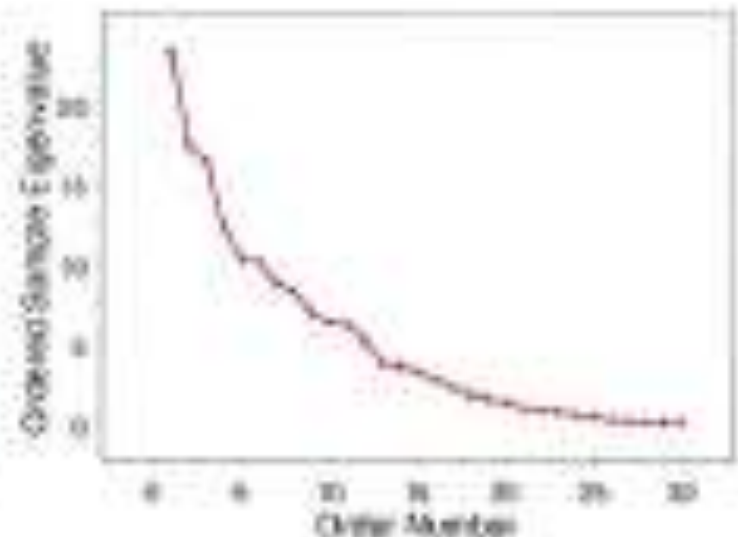
How many principal components to retain?

- **Scree Plot:** A plot of the ordered eigenvalues from largest to smallest (Catell, 1966). Inspired by the break between the mountain side and a collection of boulders usually found at its base.
- *“Elbows” are desirable, they break the largest and the smallest groups of eigenvectors.*

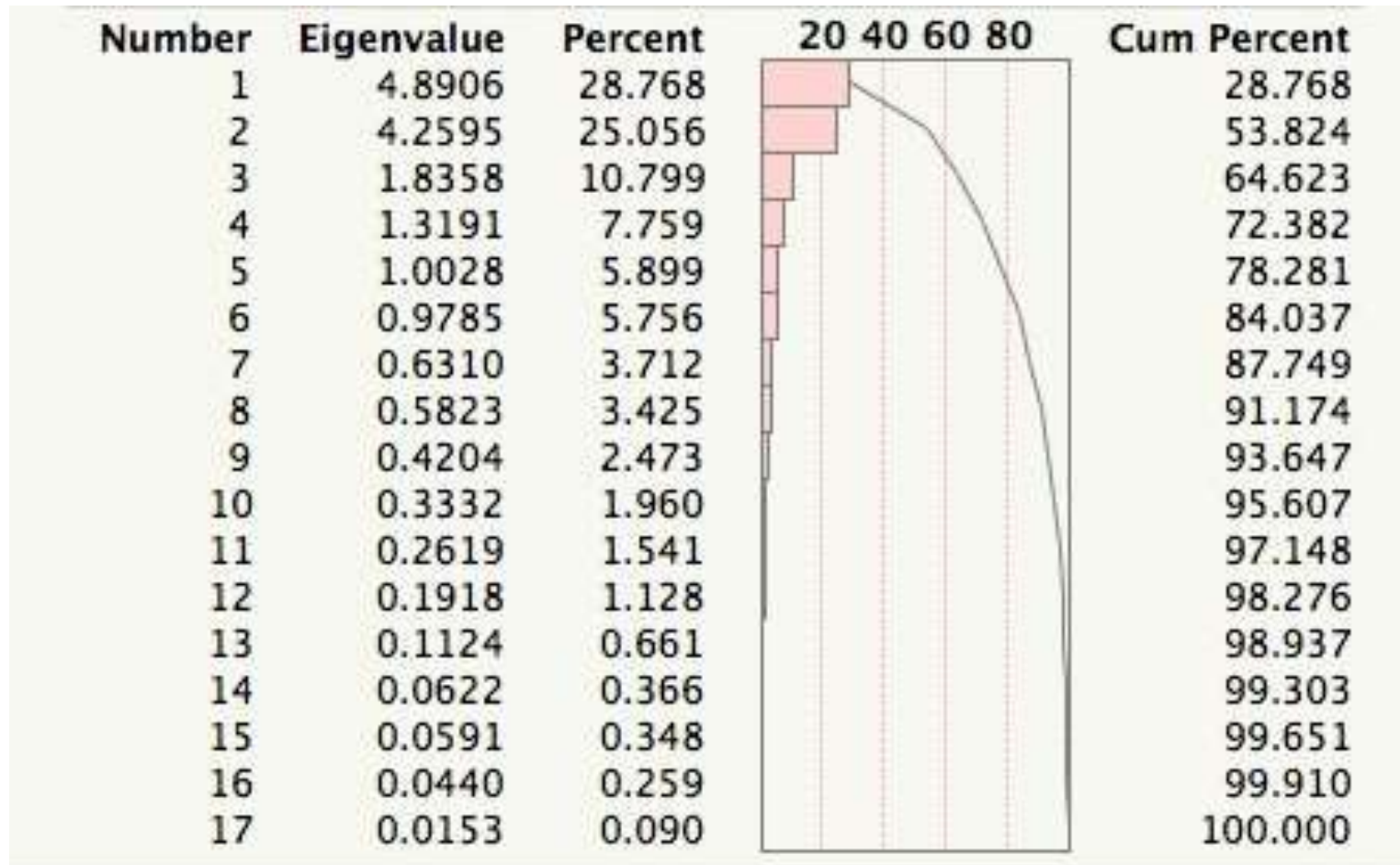
Food Example



Gaussian Simulation



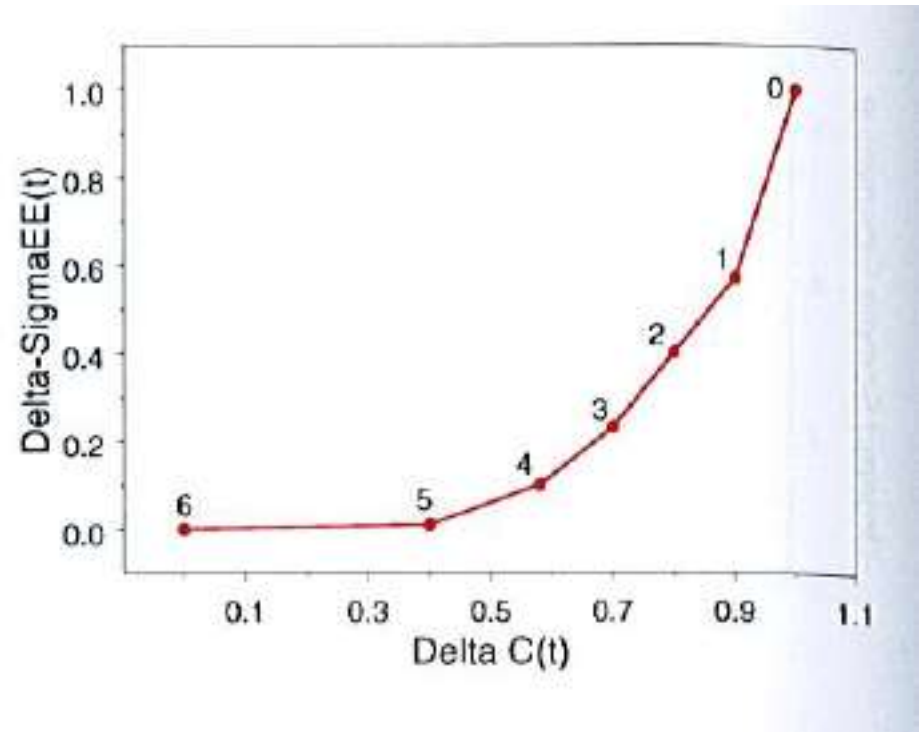
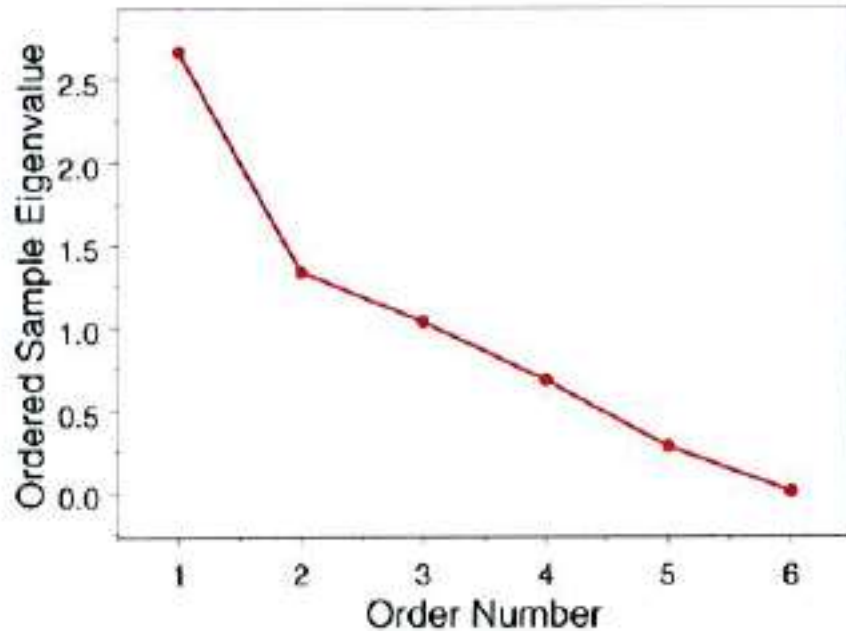
How many principal components to retain?



How many principal components to retain?

- The problem of deciding how many principal components to retain is equivalent to obtaining a useful estimate of the matrix C in the PC case. Recall: $\hat{C}^{(t)} = \hat{A}^{(t)} \hat{B}^{(t)} = \sum_{j=1}^t \hat{v}_j \hat{v}_j^T$.
- **PC Rank Trace plots:** displays the loss of information when approximating the full ranks regression by a reduced regression of increasing lengths.

How many principal components to retain?



Example: Image processing

- In the past, face recognition was primarily based upon identifying individual facial features, such as, eyes, nose, mouth etc.
- New approach: “eigen-faces” which operates directly on image-based representations of faces (Turk and Pentland, 1991).
Applications: homeland security, video surveillance, robotics, “smart” cards like passports, drivers licenses.
- Each face, as a picture image might be represented by a $(c \times d)$ -matrix of intensity values, which represent the image in 8-bit gray scale (0-255, with 0 as black and 255 as white). Images are scaled to lie in $[0,1]$.

Example: Image processing

- The matrix size ($c \times d$) depends on the degree of resolution desired.
- The matrix is then “vec’ed” by stacking the columns of the matrix under one another to form a cd -vector in an image space.

Example: an image is digitized into a (256×256) array of pixels, that face is now a point in a 65,536 dimensional space.

We can view all possible images of one particular face as a lower-dimensional manifold (face space) embedded within the high-dimensional image space.

Example: Image processing

Example from the Yale Face Database:

Contains 65 frontal face gray-scale images covering 15 individuals taken under 11 different conditions:

- Illumination – center, left, right, normal
- Expression – happy, sad, sleepy, surprised
- Glasses – with, without

Each image is of size 320×243 , which then gets stacked into an r -vector, where $r = 77,760$.

The problem is that of dimension reduction, which is the fewest number of variables necessary to identify these types of facial images.

Example: Image processing



Principal Component Analysis

Example: Image processing

Facial reconstruction:

Let $\hat{m}_X = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $C^{(t)} = A^{(t)} B^{(t)} = \sum_{j=1}^t v_j v_j^T$.

The best “rank t ” reconstruction of the i th original face can be computed as:

$$\hat{X}_i^{(t)} = \bar{X} + \hat{C}^{(t)} (X_i - \bar{X}), \quad i = 1, \dots, n.$$

Reconstruction of the i th face

Average face

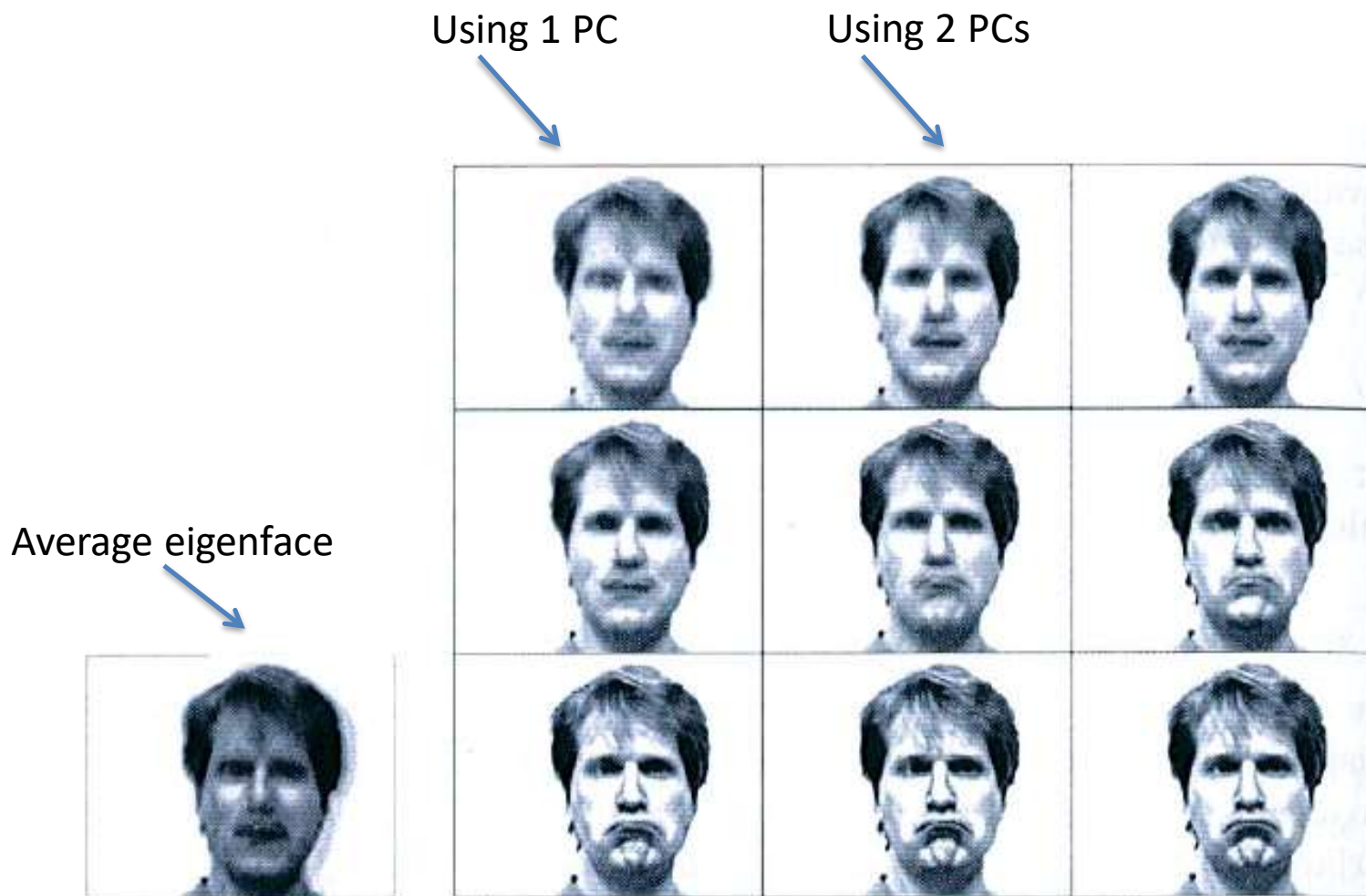


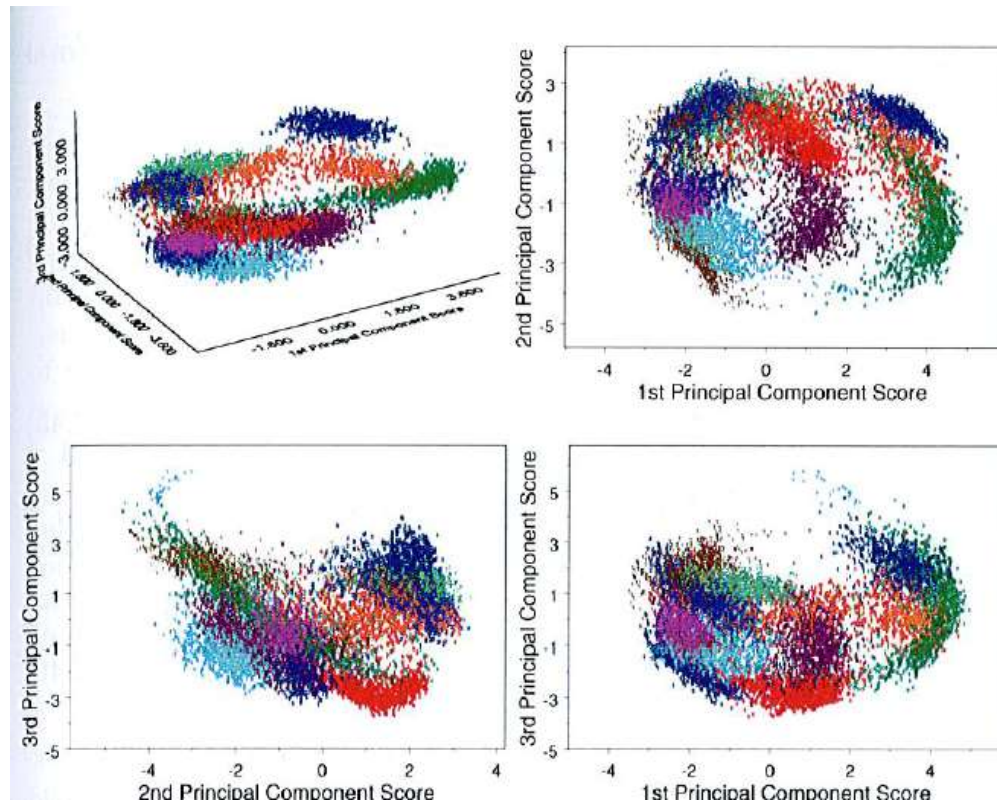
FIGURE 7.5. *The cumulative effect of the nine principal components, adding one PC at a time, for eigenface 6 (“sad”). The sad face starts to appear by the fifth PC. The average eigenface is given in the left panel.*

Example: Handwriting

- Data obtained from 44 writers.
- Each writer handwrote 250 examples of the digits 0,1,2,...,9 in a random order.
- The digits were written inside boxes of 500 x 500 pixel resolution on a pressure sensitive tablet with an integrated LCD screen.
- The raw data consists of $n=10,992$ handwritten digits of the sequence (x_t, y_t) , $t=1, 2, \dots, T$, of tablet coordinates of the pen at fixed time intervals in 100 milliseconds where x_t and y_t were integers in the range 0-500.
- Data was scaled and normalized to range between [0,100].
- Finally, each digit was sampled 8 times, giving 16 data points per digit.

Example: Handwriting

- The variances of the first five principal components are larger than unity: 4.717, 3.228, 2.557, 1.230, 1.063.
- Together these explain about 80% of the variance.



Principal Component Analysis

What can we gain from PCA?

- Depends largely on what we are trying to accomplish and the application in question.
- Depends largely on the data, and how well the variance is decomposed into orthogonal coordinates.
- Useful for “data compression”, where we pay a price for numerical accuracy for a reduction in dimensionality.
- PCA can be essential for highly correlated input variables.
- PCA can be skewed by outliers, e.g., in image processing the corruption by noisy images.